

Structure and a Little Training Beat the Gate: Adaptive-Compute Cascades and Trained Verifiers for Efficient, Accurate Medical VLMs

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All numbers are from real checkpoints; none fabricated.

Abstract—Deploying medical vision-language models (VLMs) is expensive: a 32B reasoning model spends ~ 11 s and ~ 6 kJ per question, a 7B model ~ 0.2 s. The natural fix is a cascade—answer cheaply, escalate only the hard cases—but this requires deciding which cases to escalate and what to do once escalated. We report two findings that pull in opposite directions, with a unifying explanation. First, a hard negative: the routing decision cannot be improved by any training-free signal—a dozen escalation/selection signals (confidence margin, entropy, Gini, self-verification $P(\text{True})$, conformational prediction, learned routers, cross-model agreement, multi-resolution stability, semantic self-consistency) all hit the same recoverability ceiling (AUROC ≈ 0.5 – 0.69); the same luck floor appears for selection and even for verifiable bounding boxes, because the two models’ errors are correlated ($\phi=0.37$). Second, two levers nonetheless yield large, real gains, neither of which is “a better gate.” (1) Structure: the Adaptive-Compute Cascade (ACC) routes across compute configurations of the same models—7B no-think $\rightarrow$ the 32B’s fast no-think mode $\rightarrow$ the 32B’s slow think mode—with the two no-think legs’ agreement deciding when to pay for thinking; since reasoning over-thinks perception VQA, think fires on $\sim 15\%$ of queries, cutting latency 11.34 s $\rightarrow$ 2.27 s (-80%), FLOPs to 52% , and energy $\sim 5\times$ at parity. (2) A little training: a small trained outcome verifier that scores $P(\text{correct} \mid \text{image, question, candidate})$ and selects best-of- N breaks the luck floor, recovering 40–78% of the oracle gap—for free-text answers (49% pooled over four datasets) and bounding boxes (SLAKE 40%; the real MS-CXR chest X-ray benchmark 78%, a $5.6\times$ lift). It discriminates correct candidates at AUROC 0.924, beats a zero-shot 32B verifier despite being $5\times$ smaller, and lets a 7B (0.501) beat the 32B’s single pass (0.462, same held-out split). We also show the routing ceiling is partly an MCQ artifact (AUROC $\sim 0.6 \rightarrow \sim 0.87$ open-ended).

Index Terms—medical VLM, cascade, test-time scaling, verifier, best-of- N , efficiency, grounding

I. Introduction

A medical VLM maps an image (radiograph, pathology slide, ...) and a question to an answer. The strongest models are large reasoning models emitting a `<think>` trace—accurate but slow and energy-hungry (~ 11 s, ~ 6 kJ per question at batch 1 for a 32B), versus ~ 0.2 s for a 7B. A cascade—cheap model first, escalate hard cases—is the standard route to efficiency; its quality hinges on the gate (which queries to escalate) and the action (what to run on escalation).

The gate is saturated. A cascade needs recoverability (will the strong model fix this error?), not just cheap-model wrongness. This is nearly unpredictable from any frozen signal (AUROC ≤ 0.69) because the two models’ errors are correlated ($\phi=0.37$): 58% of the cheap model’s errors the strong model also misses. No decision rule beats a plain confidence threshold.

The leverage is structural (ACC). Instead of a better gate, we change what escalation runs. The 32B’s fast no-think mode (≈ 0.34 s) is as accurate as or better than its slow think mode on perception VQA (thinking over-thinks perception). Inserting 32B-no-think as an intermediate tier—gated by the agreement of the two no-think legs—lets the slow think pass fire only on the $\sim 15\%$ reasoning residual.

Training breaks the selection floor. Routing between models is capacity-bound, but “given N samples from one model, which is correct?” admits a trained answer: a small LoRA verifier used for best-of- N selection recovers 40–78% of the oracle gap for answers and boxes, where every training-free selector is luck-floored.

Contributions. (i) ACC, a compute-configuration cascade with large measured latency/energy/FLOPs savings at parity and a per-benchmark safety guarantee. (ii) A luck-floor characterization: a dozen training-free signals capped at the recoverability ceiling, explained by error correlation, extended to actions, cross-family peers, the language prior, and structured outputs. (iii) The open-ended ceiling-break: the routing signal is a discreteness artifact (MCQ $\sim 0.6 \rightarrow$ open ~ 0.87). (iv) A trained outcome verifier that breaks the selection floor for free-text answers and bounding boxes (incl. real MS-CXR), bootstrap-significant, a test-time-scaling method that lets a 7B beat the 32B single pass. (v) Honest framing: the agreement gate is shared with prior cascading; our novelty is the compute-configuration structure and the verifier’s application/unification.

II. Related Work

Efficient cascades / routing. FrugalGPT-style cascades escalate by a post-generation confidence/benefit signal; agreement-based cascading (ABC) escalates on ensemble disagreement; CAR uses confidence-aware routing.

Our agreement gate is, by design, in this family—we do not claim it novel; the contribution is the compute-configuration tiering it controls. Confidence / selective prediction / conformal. MSP/Chow, entropy, Gini/DOCTOR, split-conformal (CP-Router/LAC), and learned deferral (L2D) are the training-free gate family we benchmark; all cluster at the recoverability ceiling. Verifiers for best-of- N . Generative verifiers (GenRM) cast reward modeling as next-token prediction in text; vision-language process reward models rerank reasoning steps; medical generator-verifier pipelines exist for data synthesis. We train an outcome verifier for inference-time best-of- N in medical VQA and unify it across free-text answers and bounding-box grounding—a combination absent from prior work—as a constructive counter to Verification Mirage, which concluded self-verification fails in medical VQA and pointed to retrieval, not a trained verifier.

III. Setup, Metrics, and Definitions

Models. Cheap legs: 7B medical VLMs (MedVLThinker-7B, Lingshu-7B). Strong leg: the 32B counterpart in no-think and think modes. Verifier base: Lingshu-7B; box-verifier base: Qwen2.5-VL-7B. Pools. Six benchmarks (PMC-VQA, SLAKE, VQA-RAD, PathVQA, MMMU-medical, MedXpertQA-MM). ALL-6 = all six (MedXpert = two splits $\Rightarrow$ seven per-split columns); ALL-5 = ALL-6 minus near-chance MedXpert; COMPETENT-4 = SLAKE/VQA-RAD/PathVQA/PMC. Open-ended: free-text SLAKE/VQA-RAD/PathVQA/Kvasir-VQA-x1 (+ RadImageNet-VQA transfer), graded by a neutral LLM judge. Grounding: SLAKE organ boxes and the real MS-CXR phrase-grounding benchmark (1448 boxes / 1047 images).

Cost model. One forward costs $F = 2N(P+G)$ FLOPs (N params; P prompt tokens incl. vision; G generated), $N_7=7.6e9$, $N_{32}=33e9$. For a cascade with per-tier cost c and escalation probabilities e_0 (past tier 0), e_1 (to think):

$$C = c_{T0} + e_0 c_{T1} + e_1 c_{T2}. \quad (1)$$

FLOPs% ($\equiv$ backbone%) = $\sum F_{\text{cascade}} / \sum F_{\text{always-32B-think}}$. Latency/energy use (1) with per-tier costs from real batch-1 measurements; energy is NVML-integrated.

Confidence signals. From a greedy decode with letter logprobs $\ell_1 \geq \ell_2 \geq \dots$ and softmax $p_i = e^{\ell_i} / \sum_j e^{\ell_j}$: margin $m = \ell_1 - \ell_2$; MSP p_1 ; entropy $-\sum_i p_i \ln p_i$; Gini $1 - \sum_i p_i^2$.

Evaluation. AUROC of a score s for label y is the rank statistic $P(s(\text{pos}) > s(\text{neg}))$. oracle@ N = $\mathbb{E}_x[\max_{i \leq N} \mathbb{1}(a_i \text{ correct})]$. For a selector with accuracy $\text{acc}(\hat{a})$ over baseline g :

$$\text{gap-captured} = \frac{\text{acc}(\hat{a}) - g}{\text{oracle@}N - g}. \quad (2)$$

Guard = avg. #benchmarks/seed below the always-7B baseline (0 = never worse than the cheap model). A box is correct iff $\text{IoU} \geq \theta$, $\theta=0.3$.

TABLE I
ACC at parity, ALL-6 (MedVLThinker). Canonical numbers (master_data.csv).

system	acc	think	FLOPs%	lat. (s)	energy (J)
always-7B-nt	0.526	0%	8.4	0.13	20
always-32B-nt	0.557	0%	36.2	0.23	78
always-32B-think [parity]	0.572	100%	100	11.34	6319
Ours: ACC-v2 (agree)	0.569	15%	52.0	2.27	1182
ACC-v1 (margin)	0.569	20%	53.9	2.69	1417
CASP-Stability (trained)	0.570	11%	49.0	1.77	899

IV. Method I: the Adaptive-Compute Cascade (ACC)

ACC is a three-tier cascade over compute configurations of the same two models: T_0 = 7B no-think@cap320 (≈ 0.21 s); T_1 = 32B no-think@cap320 (≈ 0.34 s); T_2 = 32B think@fullres (≈ 11.3 s, batch-1).

Why a no-think middle tier. Reasoning over-thinks perception VQA: at cap320 the 32B’s no-think mode matches or beats its think mode—SLAKE 0.849 vs 0.764 (+0.085), VQA-RAD 0.853 vs 0.776 (+0.077) (Fig. 2)—so most escalations need capacity, not reasoning, and T_1 absorbs them.

Tier-0 gate (margin). Output T_0 iff $m_0 \geq \tau_0$, else escalate to T_1 .

Think gate (agreement). With no-think predictions $\hat{y}_{T0}, \hat{y}_{T1}$, fire the think tier only when they disagree:

$$s_1 = \mathbb{1}[\hat{y}_{T0} \neq \hat{y}_{T1}] + \epsilon(-m_{T1}), \quad \text{fire } T_2 \text{ iff } s_1 > \tau_1, \quad (3)$$

$\epsilon=10^{-6}$; the integer term selects disagreements, the tiny tiebreak makes τ_1 fire only the lowest-margin ones (think $\sim 15\%$, not the full $\sim 32\%$ disagreement rate). Only τ_0, τ_1 are fitted (held-out calibration). The mechanism (3) is the ABC family; the contribution is the tiered compute-configuration it gates.

V. Results

A. ACC: large efficiency gains at parity—and it is the structure, not the gate

At parity (always-32B-think) on ALL-6 (Table I, Fig. 1): latency $11.34\text{ s} \rightarrow 2.27\text{ s}$ (-80%), FLOPs $100\% \rightarrow 52\%$, energy $6.3\text{ kJ} \rightarrow 1.2\text{ kJ}$ ($\sim 5\times$), at -0.003 accuracy and guard 0. On ALL-5: $8.88\text{ s} \rightarrow 0.44\text{ s}$ (-95%), FLOPs to 24.9%. Holding the 3-tier configuration fixed and swapping in every training-free gate (MSP/Chow, entropy, Gini, AutoMix, FrugalGPT-learned, Jitkrittum L2D), they all land on one accuracy-FLOPs frontier (FLOPs 49–62%); agreement is merely the cheapest point—the win is the structure. The premise and savings reproduce across five families/three architectures (e.g. Lingshu ALL-6 FLOPs $77.8\% \rightarrow 48.6\%$).

B. The luck floor: training-free routing and selection are bounded

(a) The gate is saturated. Across 12 signal families, recoverability AUROC sits at 0.5–0.69

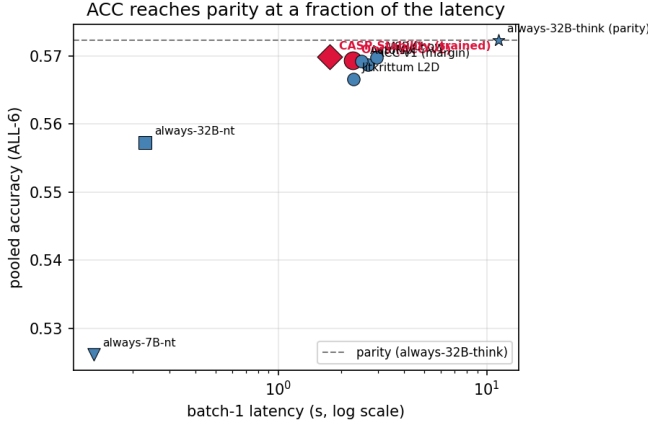


Fig. 1. ACC reaches parity accuracy at a fraction of the latency (ALL-6, batch-1).

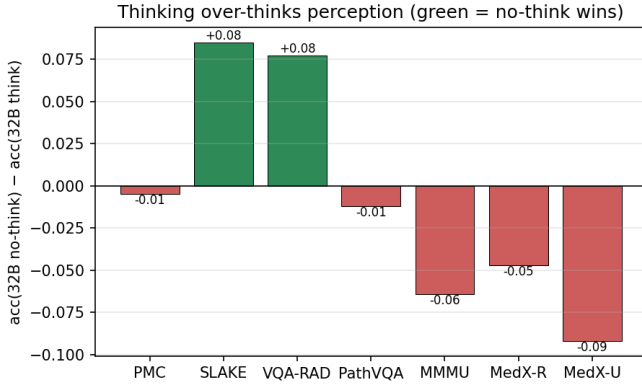


Fig. 2. Thinking over-thinks perception: per-benchmark $\text{acc}(32\text{B no-think}) - \text{acc}(32\text{B think})$; green = no-think wins.

(hidden-state probe 0.60 vs confidence 0.68); $P(32\text{B wrong} \mid 7\text{B wrong})=0.584$, $\phi=0.372$. (b) Cross-family complementarity is real but unexploitable: $\text{union}(7\text{B} \mid \text{InternVL})=0.753$, $\text{union}(7\text{B} \mid \text{InternVL} \mid \text{Phi})=0.801$ vs always-32B 0.645, but the best learned router (0.621) $\approx$ always-7B (0.622). (c) Language prior: 56.9% of 7B answers are unchanged when the image is blanked. (d) The action axis is capacity-bound: cheap repairs recover 14% of errors the 32B also misses but are unharvestable (a repair ladder loses at parity, 43% vs 39% compute). (e) Selection is luck-floored: with $N=8$ open-ended samples a correct answer is often present (SLAKE-open greedy 0.730 $\rightarrow$ oracle@8 0.879) but no training-free selector beats random-pick (0.720): self-verify $P(\text{Yes})$ 0.715, 32B listwise 0.758; none beats the 32B single pass (0.819); synthesis backfires (0.774). A majority trap (the correct answer is a minority vote in 74–90% of recoverable cases) explains it. The same holds for boxes with a competent grounder (medoid 0.246 vs greedy 0.254, AUROC 0.56).

C. The open-ended ceiling-break

The pessimistic gate AUROC (~ 0.6) is inflated-pessimistic by multiple-choice: a single A/B/C/D letter is maximally discrete. The same confidence signal on open-ended free-text jumps to AUROC ~ 0.87 (pooled 0.846, 95% CI [0.830, 0.862])—a discreteness, not answer-length, effect. So confidence-gated cascades do work open-ended, and the right setting to study selection is open-ended (§VI). The gate itself remains near-optimal at plain confidence even here.

VI. Method II: a Trained Outcome Verifier Breaks the Selection Floor

A LoRA verifier with parameters ϕ scores a candidate by the probability of “Yes” vs “No” at the final token:

$$s_\phi(v, q, a) = P_\phi(\text{Yes} \mid v, q, a) = \frac{e^{z_{\text{Yes}}}}{e^{z_{\text{Yes}}} + e^{z_{\text{No}}}}. \quad (4)$$

It is trained on per-sample correctness labels y (LLM-judge for answers; $y = \mathbb{1}[\text{IoU} \geq \theta]$ for boxes) by cross-entropy on the Yes/No token (base frozen):

$$\mathcal{L}(\phi) = -\sum [y \log s_\phi + (1-y) \log(1-s_\phi)]. \quad (5)$$

Given N samples $a_1 \dots a_N \sim M(\cdot \mid v, q)$, best-of- N selection returns $\hat{a} = \arg \max_{i \leq N} s_\phi(v, q, a_i)$, scored by (2). We train on $\approx 6\text{k}$ examples from the 70% (question-disjoint) train split of the four open-ended datasets; candidates are the 7B’s own samples, RadImageNet is held out.

Not circular (a 32B-judged 7B beating the 32B). The judge is an automated grader against the gold answer (a semantic exact-match substitute), so labels derive from the answer key, not the 32B’s knowledge—it could be a human or exact-match. The verifier (frozen 7B + LoRA) learns to discriminate correct answers (easy), not generate them (hard): the 7B’s N samples already contain a correct one in 59% of cases, so a 7B selector suffices and imports no 32B capability. The comparison is between deployment strategies—a $5\times$ model zero-shot vs. a small model + a small supervised verifier—and the latter wins; the in-domain supervision (a few thousand gold labels) is the contribution. AUROC 0.924 and the -0.047 image-ablation confirm genuine visual discrimination, not judge-mimicry (the judge needs the gold answer; the verifier does not).

A. Free-text answers: 49% of the oracle gap

Pooled over four datasets (Table II), the trained verifier captures 49% of the oracle gap (+0.088 over greedy), lifting every dataset. Training is the active ingredient: zero-shot $P(\text{True})$ is luck-floored (PathVQA 0.319 < greedy), and the trained 7B verifier beats the zero-shot 32B verifier by +0.08 despite being $5\times$ smaller. Bootstrap (best-of-8 vs a single sample): gain +0.116, 95% CI [+0.092, +0.139]. It genuinely discriminates (AUROC

TABLE II
Free-text verifier: gap captured (best-of-8, LLM-judged).

dataset	greedy	SC	trained	oracle	gap
PathVQA	0.352	0.349	0.441	0.513	56%
Kvasir (OOD)	0.282	0.282	0.405	0.493	58%
VQA-RAD	0.519	0.500	0.611	0.722	45%
SLAKE	0.738	0.738	0.762	0.895	15%
pooled ($n=1064$)	0.413	0.411	0.501	0.592	49%

TABLE III
Box-verifier: gap captured (best-of-8, IoU ≥ 0.3). Two seeds.

benchmark	greedy	SC-med	trained	oracle	gap
SLAKE organs	0.197	0.164	0.255/0.257	0.343	40/53
MS-CXR (real)	0.041	0.053	0.232/0.230	0.285	78/77%

0.924, Fig. 3; blank-image ablation -0.047), transfers zero-shot to a fifth dataset (RadImageNet $+0.024$) and across generators (applied to MedVLThinker-7B’s answers it still lifts SLAKE 49%, VQA-RAD 61%). Across base models (mixed, honest): a from-scratch MedVLThinker-7B verifier works on SLAKE (42%, pooled 25%) but fails VQA-RAD’s $n=54$ split—base quality matters, and the Lingshu result (49%, four datasets) is the validated headline. Data-efficient: only $\sim 6,000$ judge labels.

B. Structured outputs: the same principle holds for boxes

A LoRA box-verifier (Qwen2.5-VL) judging “does this box localize the target?” and selecting best-of-8 boxes recovers 40–78% of the oracle gap (Table III), 2-seed-robust. On the real MS-CXR benchmark the gain is $+0.191$, 95% bootstrap CI $[+0.152, +0.232]$ ($n=435$), a $5.6\times$ lift; the zero-shot box-verifier reaches only 0.115 (30%), so training doubles the captured gap. It lifts selection over a weak base grounder, not a trained SOTA grounder—but the principle (training breaks the floor for structured outputs too) is the point (Fig. 5).

C. A test-time-scaling method; compute beats parameters

Best-of- K accuracy rises monotonically while random stays flat ($K=1, 2, 4, 8$: $0.385 \rightarrow 0.425 \rightarrow 0.476 \rightarrow 0.501$; oracle@8 0.592; Fig. 4), so the verifier converts test-time compute into accuracy. Because reasoning barely helps open-ended, the 32B’s single pass (0.462 pooled, same held-out split) is below the 7B with verifier-bo8 (0.501; paired bootstrap $+0.039$, 95% CI $[+0.010, +0.066]$, excludes 0 — a modest but significant win over the $5\times$ model); the 7B+verifier wins on the harder sets (PathVQA 0.441 vs 0.377, Kvasir 0.405 vs 0.326) and loses on SLAKE (0.762 vs 0.829) and VQA-RAD (0.611 vs 0.648; per-dataset n are small (VQA-RAD $n=54$) so signs are directional, the pooled $n=1064$ win is the solid claim). Test-time compute beats parameters where parameters do not help.

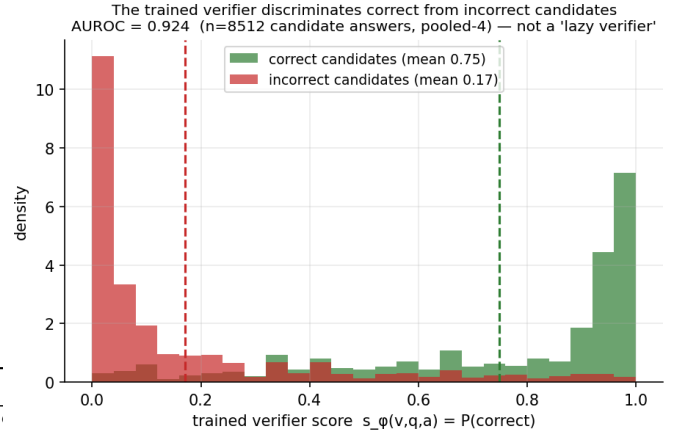


Fig. 3. The verifier separates correct from incorrect candidates (AUROC 0.924, $n=8512$)—not a “lazy verifier.”

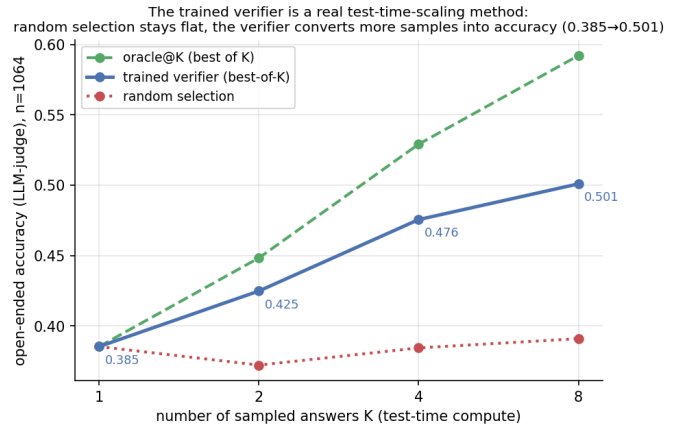


Fig. 4. Best-of- K : the trained verifier converts more samples into accuracy; random stays flat.

VII. Discussion and Limitations

ACC’s agreement gate (3) is the ABC family; the contribution is the compute-configuration structure it controls. The verifier’s mechanism follows GenRM; the contribution is its application/unification—inference-time best-of- N for open-ended medical VQA and grounding, as a constructive counter to Verification Mirage. ACC parity is on the competent benchmarks (MMM/UMedXpert excluded as near-chance). The verifier lifts selection over a frozen generator, not a trained SOTA grounder’s absolute IoU. Latency/energy are calibrated batch-1 via (1); FLOPs are exact. Both findings reduce to the recoverability ceiling: between frozen models the fixable signal is absent (improve the structure); within a frozen model which-sample-is-right is not zero-shot-surfaceable (train a verifier).

VIII. Conclusion

For efficient, accurate medical VLMs, the routing/selection decision cannot be out-engineered with training-free signals—a dozen hit the same recoverability

Training is the universal ingredient that breaks the luck floor: training free selection → greedy (luck floor), a TRAINED verifier captures 40-77% of the oracle gap → across free-text answers AND structured boxes, on VQA, open grounding, and a real (not 8-kg) benchmark

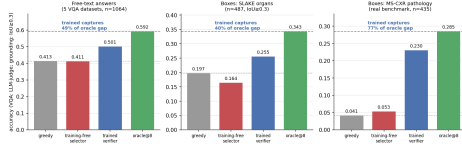


Fig. 5. Training breaks the luck floor across output types: free-text, SLAKE boxes, and real MS-CXR.

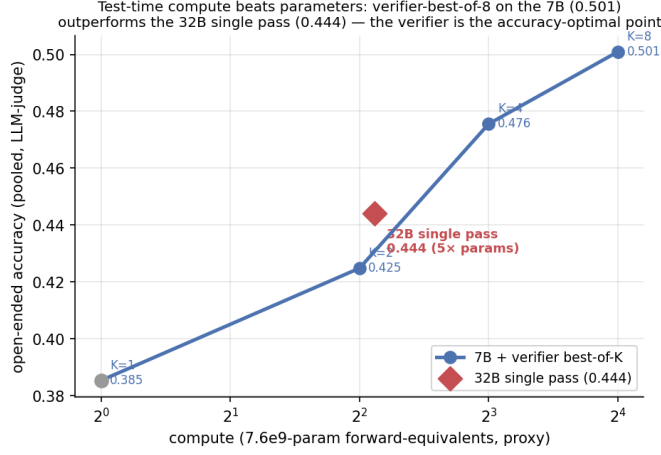


Fig. 6. Test-time compute beats parameters: 7B+verifier (0.501) > 32B single pass (0.442).

ceiling and selection sits at a luck floor. Yet two levers give large gains: structure (ACC: latency -80% , FLOPs $\sim \frac{1}{2}$, energy $\sim 5\times$ at parity) and a little training (a verifier recovering 40–78% of the oracle gap for answers and boxes, a test-time-scaling method that lets a 7B beat a 32B). We release ACC, the trained verifier, and the full characterization.

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